Advanced Computer Networks

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Markov Process: Example problems

DSSRG: Decentralized Smart Systems Research Group

https://sites.google.com/iitbbs.ac.in/dssrg
Or Google dssrg iitbbs

Questions Set 1 (Marks 30)

- Consider a **Markov Chain** with the following transition probability matrix:
-

$$
P = \begin{array}{c} \\ \\ \\ \\ \end{array}
\begin{array}{c}
\begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\
\begin{bmatrix}
1/4 & 1/4 & 1/4 & 1/4 \\
0 & 0 & 1 & 0 \\
0 & 0 & 0 & 1 \\
1 & 0 & 0 & 0
\end{bmatrix}
\end{array}
\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array}
$$

Questions Set 1 (Marks 30)

1. Draw the state transition diagram - the states are numbered as 1, 2, 3 and 4 (Marks 10)

2. Given that the system is currently in **state 1**, what is the **probability** that it will be **state 3** in **4 transitions** ?

(Marks 10)

3. What is the **limiting (steady) state probabilities** π_1, π_2, π_3 and π_4. (Marks 20)

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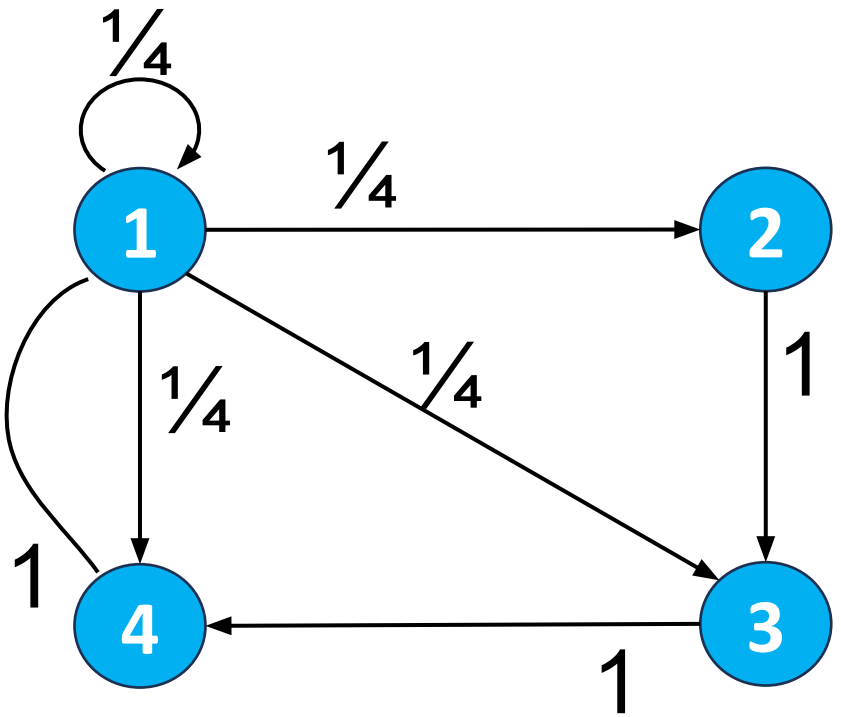
Solutions 1

- Ans. 1



1/4	1/4	1/4	1/4
0	0	1	0
0	0	0	1
1	0	0	0

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(Marks 10)

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Solutions 1

P =

¼	¼	¼	¼
0	0	1	0
0	0	0	1
1	0	0	0

$P^2 = P \times P$

¼	¼	¼	¼
0	0	1	0
0	0	0	1
1	0	0	0

¼	¼	¼	¼
0	0	1	0
0	0	0	1
1	0	0	0

=

0.3125	0.06	0.3125	0.3125
0	0	0	1
1	0	0	0
0.25	0.25	0.25	0.25

Solutions 1

$P^2 =$

0.3125	0.06	0.3125	0.3125
0	0	0	1
1	0	0	0
0.25	0.25	0.25	0.25

$P^3 = P^2 \times P$

0.3125	0.06	0.3125	0.3125
0	0	0	1
1	0	0	0
0.25	0.25	0.25	0.25

¼	¼	¼	¼
0	0	1	0
0	0	0	1
1	0	0	0

=

0.3906	0.0781	0.1406	0.3906
1	0	0	0
0.25	0.2	0.25	0.25
0.3125	0.0625	0.3125	0.3125

Solutions 1

$P^3 =$

0.3906	0.0781	0.1406	0.3906
1	0	0	0
0.25	0.2	0.25	0.25
0.3125	0.0625	0.3125	0.3125

$P^4 = P^3 \times P$

0.3906	0.0781	0.1406	0.3906
1	0	0	0
0.25	0.25	0.25	0.25
0.3125	0.0625	0.3125	0.3125

¼	¼	¼	¼
0	0	1	0
0	0	0	1
1	0	0	0

=

0.4882	0.0976	**0.1757**	0.2382
0.25	0.25	0.25	0.25
0.3125	0.0625	0.3125	0.3125
0.3906	0.0781	0.1406	0.3906

$$P^4_{13} = P(X_4 = 3 \mid X_0 = 1) = \mathbf{45/256 = 0.1757}$$

Questions Set 1 (Marks 30)

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(Marks 10)

3. What is the **limiting (steady) state probabilities** π_1, π_2, π_3 and π_4. (Marks 20)

Solutions 1

- Ans. 3

Limiting probabilities $\pi = [\ \pi_1\ ,\ \pi_2\ ,\ \pi_3\ ,\ \pi_4\ ]$ satisfy $\pi P = \pi$ and $\sum \pi_i = 1$.

From $\pi P = \pi$,

Equation for State 1

State 1 can be entered from:

- State 1 with prob 1/4
- State 4 with prob 1

$$\pi_1 = \tfrac{1}{4}\pi_1 + \pi_4$$

$$\pi_4 = \tfrac{3}{4}\pi_1$$

Equation for State 2

$$\pi_2 = \tfrac{1}{4}\pi_1$$

Equation for State 3

$$\pi_3 = \tfrac{1}{4}\pi_1 + \pi_2$$

$$\pi_3 = \tfrac{1}{2}\pi_1$$

Equation for State 4

$$\pi_4 = \tfrac{1}{4}\pi_1 + \pi_3$$

$$\pi_4 = \tfrac{3}{4}\pi_1$$

Solutions 1

- 3 Continue..

$$\pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$$

Substitute everything in terms of π_1

$$\pi_1 + \tfrac{1}{4}\pi_1 + \tfrac{1}{2}\pi_1 + \tfrac{3}{4}\pi_1$$

$$\pi_1 = (1/2.5) = \tfrac{2}{5}$$

Compute remaining probabilities

$$\pi = \left(\tfrac{2}{5}, \tfrac{1}{10}, \tfrac{1}{5}, \tfrac{3}{10}\right)$$

Question Set 2 (30 pts)

- A professor for a course left **3 books** - Labelled 1, 2 and 3 - in the reference section of the library. Students are **not allowed** to leave the library with these books. However, Bob has discovered a way to beat the system:
- If he comes **just before the library closes**, then he can check out one of the books for the night and return it the following morning.
- He is only allowed to check out one book at a time.

- Bob's habit for checking out the books can be characterized as follows:

1. Given that he checks out **book 1** today, he is equally likely to check out **books 1 or 2** tomorrow.
2. Given that he checks out **book 2** today, tomorrow he will check out **book 1 with probability 0.5**, **book 2 with probability 0.25** and **book 3 with probability 0.25**.
3. Given that he checks out **book 3** today, tomorrow he will check out **book 2 with probability 0.25** and **book 3 with probability 0.75**.

Questions Set 2

1. Draw the state transition diagram (the states are to be numbered as 1, 2 and 3).
(Marks 10)

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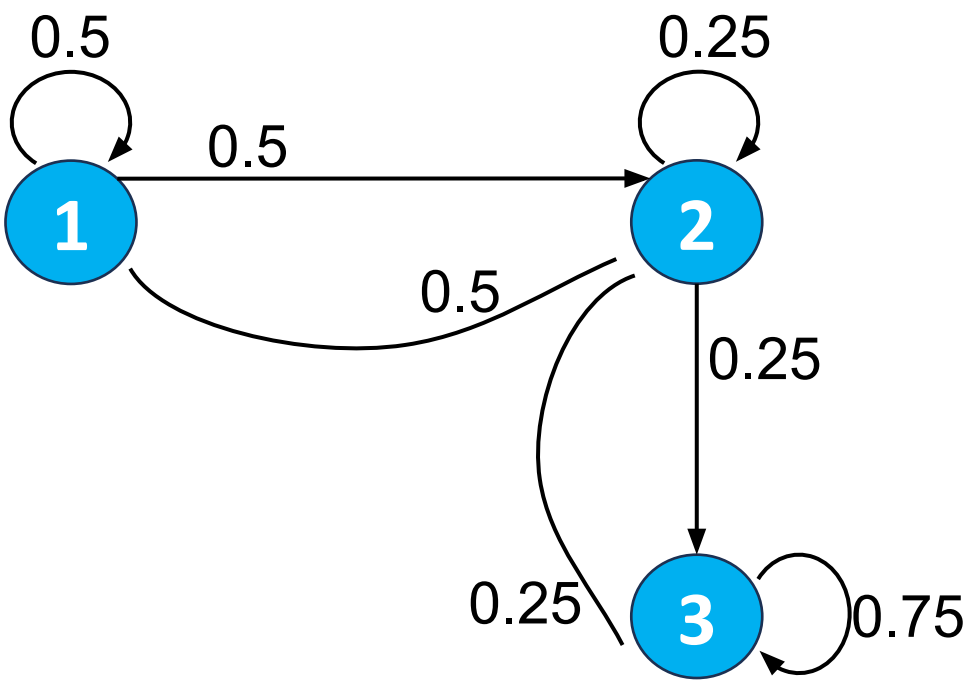
(Marks 10)

- Ans.



2. Give the **one-step probability matrix** (Marks 10)

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Probability matrix P

P =

0.5	0.5	0
0.5	0.25	0.25
0	0.25	0.75

3. Given that Bob has checked out book 1 today, what is the probability that he will check out book 3 in 4 days time ?

(Marks 10)

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(Marks 10)

P =

0.5	0.5	0
0.5	0.25	0.25
0	0.25	0.75

P^2 =

0.5	0.5	0
0.5	0.25	0.25
0	0.25	0.75

0.5	0.5	0
0.5	0.25	0.25
0	0.25	0.75

=

0.5	0.375	0.125
0.375	0.375	0.25
0.125	0.25	0.625

$P^2 =$

0.5	0.375	0.125
0.375	0.375	0.25
0.125	0.25	0.625

$P^3 = P^2 \times P$

0.5	0.375	0.125
0.375	0.375	0.25
0.125	0.25	0.625

0.5	0.5	0
0.5	0.25	0.25
0	0.25	0.75

=

0.4375	0.375	0.1875
0.375	0.3473	0.2812
0.1875	0.2812	0.5312

$P^3 =$

0.4375	0.375	0.1875
0.375	0.3473	0.2812
0.1875	0.2812	0.5312

$P^4 = P^3 \times P$

0.4375	0.375	0.1875
0.375	0.3473	0.2812
0.1875	0.2812	0.5312

0.5	0.5	0
0.5	0.25	0.25
0	0.25	0.75

=

0.4062	0.3593	**0.2343**
0.3593	0.3437	0.2968
0.2343	0.2968	0.4687

$$P(X_4 = 3 \mid X_0 = 1) = \mathbf{0.234375}$$

4. Find the limiting (steady) state probabilities π1, π2 and π3. (Marks 20)

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Solve $\pi P = \pi$ and $\pi_1 + \pi_2 + \pi_3 = 1$.

Equations 1. $\pi_1 = 0.5\pi_1 + 0.5\pi_2$

. 2. $\pi_2 = 0.5\pi_1 + 0.25\pi_2 + 0.25\pi_3$

3. $\pi_3 = 0.25\pi_2 + 0.75\pi_3$

From 1, $0.5\pi_1 = 0.5\pi_2 \Longrightarrow \pi_1 = \pi_2$

From 3, $\pi_3 - 0.75\pi_3 = 0.25\pi_2 \Longrightarrow 0.25\pi_3 = 0.25\pi_2 \Longrightarrow \pi_3 = \pi_2$

So, $\pi_1 = \pi_2 = \pi_3$

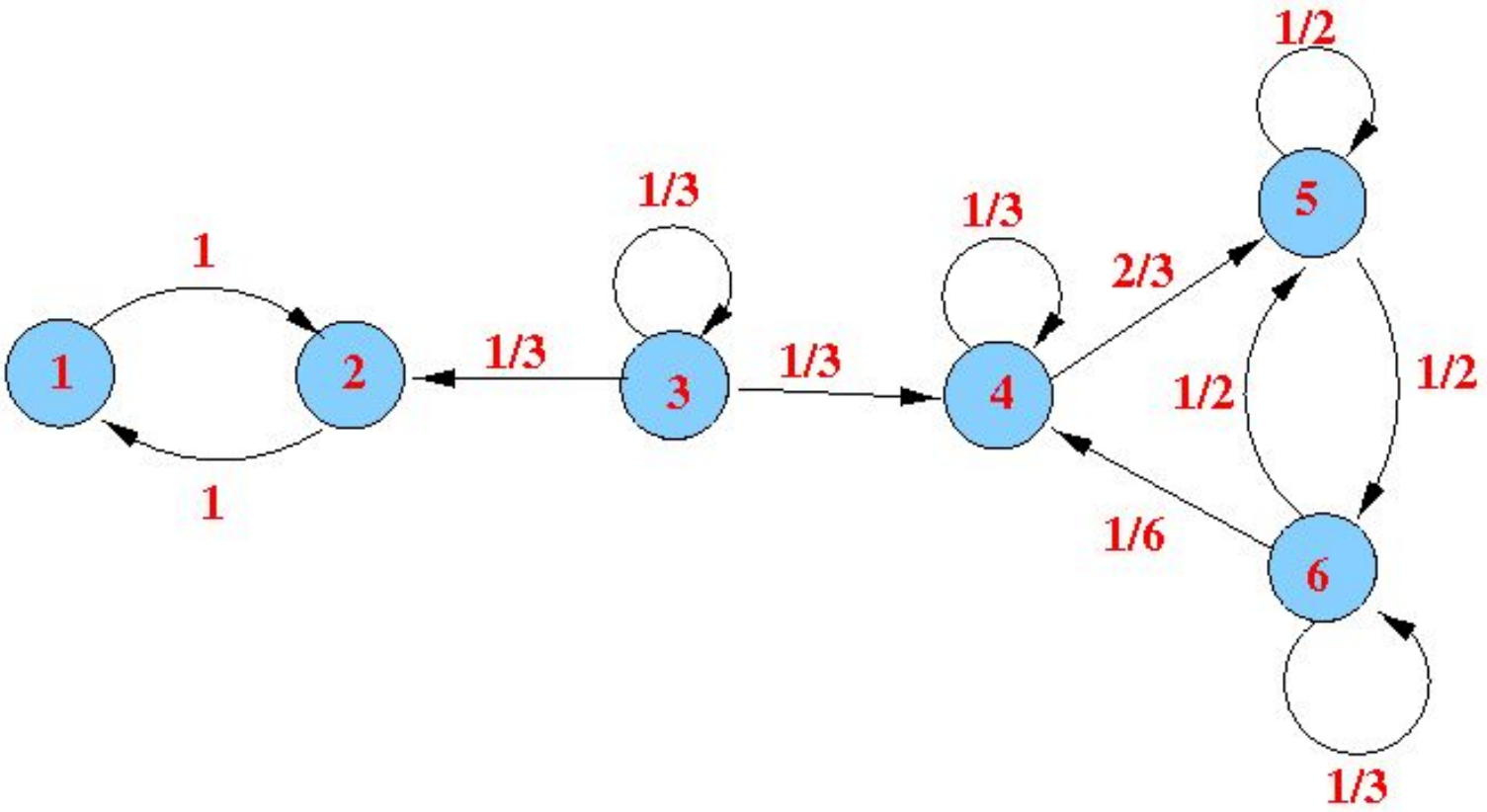
Sum = $1 \Rightarrow 3\pi_1 = 1 \Longrightarrow \pi_1 = ⅓$

Thus $\pi_1 = \pi_2 = \pi_3 = ⅓$

$\pi_1 = ⅓, \pi_2 = ⅓, \pi_3 = ⅓$

Question 3 (40 pts)

- Consider the following discrete-time Markov chain:



1. Give the **one-step transition probability matrix** (10 pts)

1. Give the **one-step transition probability matrix** (10 pts)

P =

0	1	0	0	0	0
1	0	0	0	0	0
0	1/3	1/3	1/3	0	0
0	0	0	1/3	2/3	0
0	0	0	0	1/2	1/2
0	0	0	1/6	1/2	1/3

2. List all transient states

2. List all transient states

Transient state:

A transient state of a Markov chain is a state where the stationary probability is equal to zero (0)

2. List all transient states

P =

0	1	0	0	0	0
1	0	0	0	0	0
0	1/3	1/3	1/3	0	0
0	0	0	1/3	2/3	0
0	0	0	0	1/2	1/2
0	0	0	1/6	1/2	1/3

P^2 =

1	0	0	0	0	0
0	1	0	0	0	0
0.333	0.111	0.111	0.222	0.222	0
0	0	0	0.111	0.556	0.333
0	0	0	0.083	0.500	0.417
0	0	0	0.111	0.528	0.361

$P^4 =$

1	0	0	0	0	0
0	1	0	0	0	0
0.370	0.123	0.012	0.068	0.259	0.167
0	0	0	0.096	0.515	0.389
0	0	0	0.097	0.516	0.387
0	0	0	0.096	0.516	0.387

$P^8 =$

1	0	0	0	0	0
0	1	0	0	0	0
0.375	0.125	0	0.049	0.258	0.193
0	0	0	0.097	0.516	0.387
0	0	0	0.097	0.516	0.387
0	0	0	0.097	0.516	0.387

Transient State/s: 3

3. List all Recurrent States

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Recurrent state:

A recurrent state of a Markov chain is a non-transient state (stationary probability nonzero)

3. List all Recurrent States

$P^8 =$

1	0	0	0	0	0
0	1	0	0	0	0
0.375	0.125	0	0.049	0.258	0.193
0	0	0	0.097	0.516	0.387
0	0	0	0.097	0.516	0.387
0	0	0	0.097	0.516	0.387

Recurrent States: 1, 2, 4, 5, 6

4. List all periodic states

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Periodic states:

A periodic state is a special subclass of recurrent states

A periodic state is a recurrent state say with a period d if it re-occurs after exactly d steps

4. List all periodic states

$P^8 =$

1	0	0	0	0	0
0	1	0	0	0	0
0.375	0.125	0	0.049	0.258	0.193
0	0	0	0.097	0.516	0.387
0	0	0	0.097	0.516	0.387
0	0	0	0.097	0.516	0.387

Periodic States: 1,2

4. List all periodic states

For a state i,

$$\boxed{d(i) = \gcd\{\, n \geq 1 \;:\; (P^n)_{ii} > 0 \,\}}$$

Where:

- P = transition matrix
- P^n = matrix after n steps
- $(P^n)_{ii}$ = probability of coming **back to state** i in n steps

5. List all chains

5. List all chains

A chain (communicating class) in a Markov process is a maximal set of states where every state can be reached from every other state (mutual reachability).

5. List all chains

Find the adjacency matrix A from the state transition diagram

A =

0	1	0	0	0	0
1	0	0	0	0	0
0	1	1	1	0	0
0	0	0	1	1	0
0	0	0	0	1	1
0	0	0	1	1	1

5. List all chains

Find the matrix A^2 from the matrix A.

$A^2 =$

1	0	0	0	0	0
0	1	0	0	0	0
1	1	1	1	1	0
0	0	0	1	1	1
0	0	0	1	1	1
0	0	0	1	1	1

5. List all chains

Find the matrix A^3 from the matrix A.

$A^3 =$

0	1	0	0	0	0
1	0	0	0	0	0
1	1	1	1	1	1
0	0	0	1	1	1
0	0	0	1	1	1
0	0	0	1	1	1

5. List all chains

Find the matrix A^4 from the matrix A.

$A^4 =$

1	0	0	0	0	0
0	1	0	0	0	0
1	1	1	1	1	1
0	0	0	1	1	1
0	0	0	1	1	1
0	0	0	1	1	1

5. List all chains

Find the matrix A^5 from the matrix A.

$A^5 =$

0	1	0	0	0	0
1	0	0	0	0	0
1	1	1	1	1	1
0	0	0	1	1	1
0	0	0	1	1	1
0	0	0	1	1	1

5. List all chains

Then find the reachability matrix R.

$R = I + A^1 + A^2 + A^3 + A^4 + A^5$

Use Boolean addition.

R =

1	1	0	0	0	0
1	1	0	0	0	0
1	1	1	1	1	1
0	0	0	1	1	1
0	0	0	1	1	1
0	0	0	1	1	1

5. List all chains

Rule:

If $R_{ij} > 0$, then state j is reachable from state i.

If $R_{ij} > 0$ and $R_{ji} > 0$, then i and j communicate with each other and they both lie within same chain.

R =

1	1	0	0	0	0
1	1	0	0	0	0
1	1	1	1	1	1
0	0	0	1	1	1
0	0	0	1	1	1
0	0	0	1	1	1

5. List all chains

R =

1	1	0	0	0	0
1	1	0	0	0	0
1	1	1	1	1	1
0	0	0	1	1	1
0	0	0	1	1	1
0	0	0	1	1	1

Answer: {1, 2}, {3}, {4, 5, 6}

6. Given that the process is currently in state 3, what is the probability that it will never enter state 1 (10 pts)

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$P^8 =$

1	0	0	0	0	0
0	1	0	0	0	0
0.375	0.125	0	0.049	0.258	0.193
0	0	0	0.097	0.516	0.387
0	0	0	0.097	0.516	0.387
0	0	0	0.097	0.516	0.387

Ans: 1-0.375 = 0.625

Questions:

7. Given that the process starts in **state 4**. We let the process run for a very long time (say 1,000,000 steps) and then make an observation. What is the probability that the process is in **state 6** when we observe it ? (10 pts)

Solutions 1

- Ans. 3

Limiting probabilities $\pi = [\ \pi_4\ ,\ \pi_5\ ,\ \pi_6\ ]$ satisfy $\pi P = \pi$ and $\sum \pi_i = 1$.

From $\pi P = \pi$,

Equation for State 4

$\pi_4 = \frac{1}{3}\pi_3 + \frac{1}{3}\pi_4 + \frac{1}{6}\pi_4$

$\pi_6 = 4\pi_4$

Equation for State 5

$\pi_5 = \frac{1}{2}\pi_5 + \frac{1}{2}\pi_6 + \frac{2}{3}\pi_4$

$\pi_5 = (16/3)\pi_4$

Equation for State 6

$\pi_6 = \frac{1}{3}\pi_6 + \frac{1}{2}\pi_5$

$\pi_4 + \pi_5 + \pi_6 = 1$

$\pi_4 = 3/31$

$\pi_5 = 16/31$

$\pi_6 = 12/31$

$\pi_4 = 3/31$

$\pi_5 = 16/31$

$\pi_6 = 12/31$